

NETWORKS AND COMPLEXITY

Solution 22-1

*This is an example solution from the forthcoming book *Networks and Complexity*.
Find more exercises at <https://github.com/NC-Book/NCB>*

Ex 22.1: Kronecker products [Lvl. 1]

Consider the matrices

$$\mathbf{A} = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \quad \mathbf{B} = \begin{pmatrix} 4 \end{pmatrix} \quad \mathbf{C} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \quad (1)$$

Compute a) $\mathbf{A} \otimes \mathbf{B}$, b) $\mathbf{A} \otimes \mathbf{C}$ and c) $\mathbf{C} \otimes \mathbf{A}$

Solution

$$\mathbf{A} \otimes \mathbf{B} = \begin{pmatrix} 4 & 8 \\ 12 & 16 \end{pmatrix} \quad (2)$$

$$\mathbf{A} \otimes \mathbf{C} = \begin{pmatrix} 1 & 0 & 2 & 0 \\ 0 & 0 & 0 & 0 \\ 3 & 0 & 4 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \quad (3)$$

$$\mathbf{C} \otimes \mathbf{A} = \begin{pmatrix} 1 & 2 & 0 & 0 \\ 3 & 4 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \quad (4)$$